

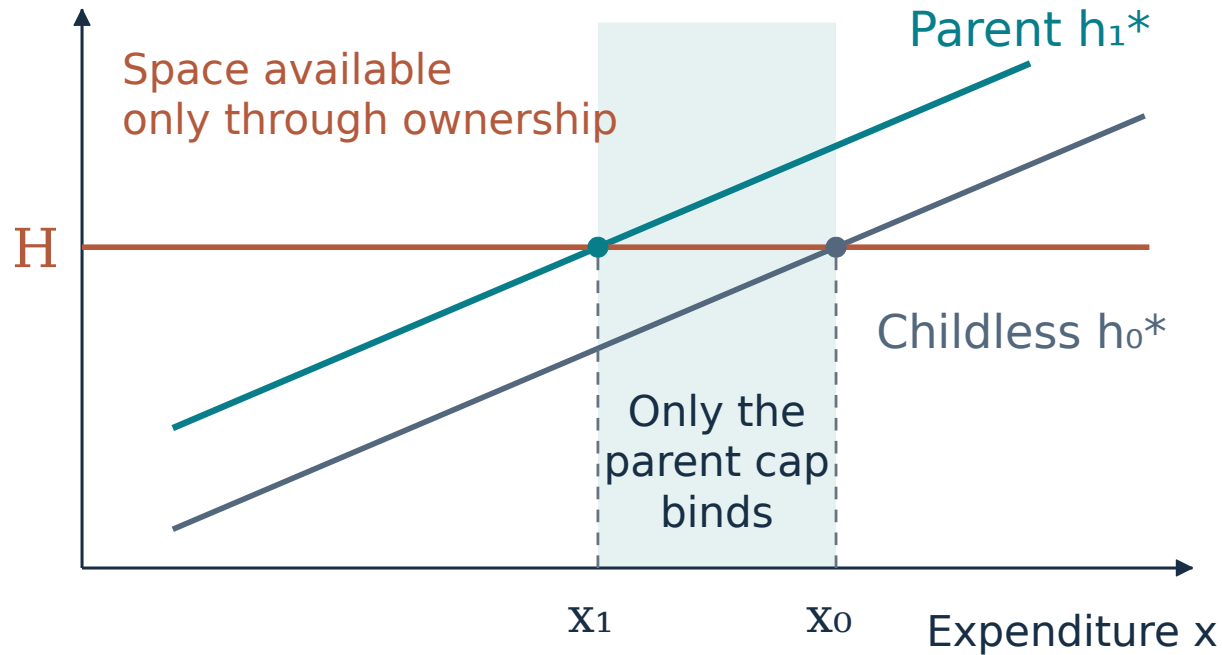
Tenure segmentation and family space

Rental budget: $c + rh = x$, with $h \leq H$. Larger homes require ownership.

$$h_1^*(x) = h_0^*(x) + \alpha h_p$$

α : consumption share. h_p : parenthood space floor.

Desired rooms



Benchmark expenditure thresholds: $x_1 \approx 2.160$, $x_0 \approx 3.157$

Parents reach the rental cap first

$$x_1 = r(H - \alpha h_p)/(1 - \alpha)$$

$$x_0 = rH/(1 - \alpha) > x_1$$

When $x_1 < x < x_0$, the childless rental option stays unconstrained.

$$\partial D / \partial H = \zeta_1 - \zeta_0 > 0$$

D : birth advantage. ζ_m : value of relaxing the cap.

Conditional on renting and common x

Ownership finance: $c + k(h) + (1 - \phi)Ph \leq M$

M includes income and liquid wealth. P : price per room. ϕ : LTV. $k(h)$: other current costs.

Rental limits can amplify the local cost of risk

Commit to the child before future expenditure X is realized.
Compare capped and uncapped rental options near x_1 .

$$L(x) = D(\infty, x) - D(H, x)$$

$$L(x_1) = L'(x_1) = 0$$

$$L''(x_1^+) = [(1 - \alpha)/\alpha] \lambda / (x_1 - rh_p) > 0$$

λ is the marginal value of expenditure at x_1 .

$X = x_1 \pm \varepsilon$, each with probability $1/2$

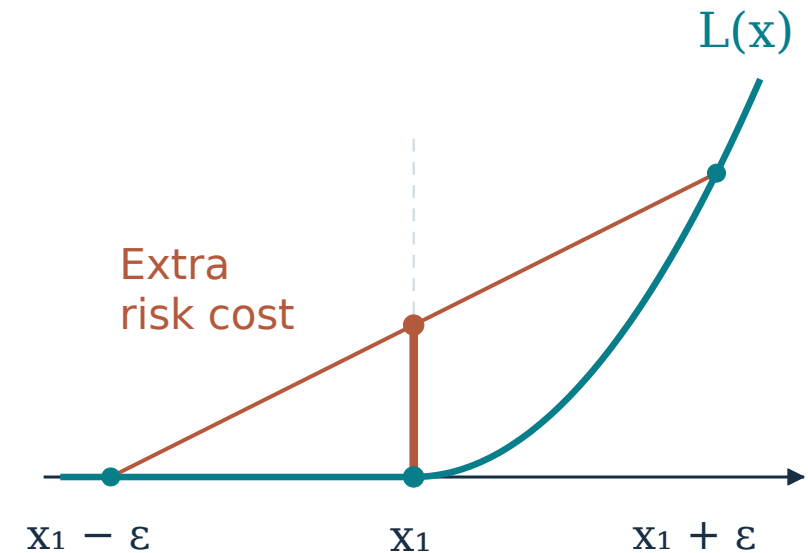
$$\text{Extra penalty} = \frac{1}{4} L''(x_1^+) \varepsilon^2 + o(\varepsilon^2) > 0$$

No transaction cost is needed for this local result.

Scope: the rental option remains relevant. Saving and tenure choice can change the sign.

Test: rental expansion across future risk levels at common states and expected earnings, including zero selling costs.

Cap loss near x_1



$$E[L(X)] > L(E[X]) = 0$$

Local quadratic approximation